

Computer Architectures

Exam of 01/07/2025

B2

Question 1 (6 points)

A linear congruential generator (LCG) is an algorithm that produces a sequence of pseudo-random numbers. Write the `LCGsequence` subroutine, in ARM assembly, that returns the n -th element of a sequence, computed as follows:

$$X_n = ((aX_{n-1} + c) \text{ EXOR } (X_{n-1} \gg s)) \bmod m$$

where $\gg s$ indicates the operation of logical shift right by s positions.

The subroutine receives in input (in the specified order):

1. the previous element of the sequence X_{n-1}
2. the multiplier a
3. the increment c
4. the shift s
5. the module m

In order to test the procedure, define an array of bytes with DIM elements in a readwrite area of memory (DIM is a constant that can be freely chosen). Then, write a loop of DIM iterations in the `Reset_Handler` to fill the array. In each iteration of the loop, you call the `LCGsequence` subroutine with the appropriate parameters.

Example. With the seed 6, $a = 157$, $c = 3$, $s = 3$, $m = 256$, the first 10 elements are:

iteration	value1 = $aX_{n-1} + c$	value2 = $X_{n-1} \gg s$	value1 EXOR value2	$X_n = (\text{value1 EXOR value2}) \bmod m$	Hex
0	$157 * 6 + 3 = 945$	0	$945 \text{ EXOR } 0 = 945$	$945 \bmod 256 = 177$	B1
1	$157 * 177 + 3 = 27792$	22	$27792 \text{ EXOR } 22 = 27782$	$27782 \bmod 256 = 134$	86
2	21041	16	21025	33	21
3	5184	4	5188	68	44
4	10679	8	10687	191	BF
5	29990	23	30001	49	31
6	7696	6	7702	22	16
7	3457	2	3459	131	83
8	20570	16	20554	74	4A
9	11621	9	11628	108	6C

Question 2 (6 points)

Extend the project to control leds 4-7 based on the values generated by the LCG.

First, start timer 1 in order to generate an interrupt every 2.5 seconds.

Then, in the handler of timer 1, call the `LCGsequence` subroutine to obtain the n -th element of the sequence. Since the value ranges in $[0, 255]$, compute the modulo 4 of such value:

- if $\text{value} \bmod 4 == 0$, switch on led 4
- if $\text{value} \bmod 4 == 1$, switch on led 5
- if $\text{value} \bmod 4 == 2$, switch on led 6
- if $\text{value} \bmod 4 == 3$, switch on led 7

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Use the same parameters as in the example: seed 6, $a = 157$, $c = 3$, $s = 3$, $m = 256$. The LCGsequence subroutine is called only 10 times (i.e., during the first 10 interrupts of timer 1). The leds are switched on in the following order:

iteration n	X_n	$X_n \bmod 4$	Led on
0	177	1	5
1	134	2	6
2	33	1	5
3	68	0	4
4	191	3	7
5	49	1	5
6	22	2	6
7	131	3	7
8	74	2	6
9	108	0	4

Important: Ensure that only one led is on at any given time. Before switching on a new led, turn off the previously lit led (from the previous timer interrupt).

Question 3 (6 points)

Extend the project to implement a rhythm game using the joystick on the board. The game will involve moving the joystick in response to the leds being switched on.

When a led lights up, the player must move the joystick accordingly:

- led 4: move the joystick up
- led 5: move the joystick left
- led 6: move the joystick right
- led 7: move the joystick down

Only the first movement of the joystick has to be considered (further movements within the 2.5-second interval are ignored). If the movement corresponds to the correct direction for the lit led, increment the variable `hit`, otherwise increment `miss`. Regardless of the correctness, after the joystick movement the led is immediately switched off.

The game continues for 10 sequence elements. After 10 elements, the game ends and the player's score is evaluated. If `hit > miss`, the player wins: turn on led 10 to indicate victory. Otherwise, the player loses: turn on led 11 to indicate defeat.